DIEGO OLIVEIRA DA CRUZ

BACKEND ENGINEER · APPLIED AI

Turning business problems into reliable backends.

Penápolis, SP, Brazil | doc89@outlook.com.br | github.com/DiegoOCruz

PROFESSIONAL SUMMARY

Software engineering professional with a law degree and 15+ years of experience at the intersection of business operations, requirements, risk analysis, and process improvement. Focused on backend development and applied AI, with hands-on projects using Java, Spring Boot, Python, PostgreSQL/pgvector, Ollama, and retrieval-augmented generation (RAG). Combines software engineering practices with the ability to understand business problems, model domain rules, and turn requirements into reliable technical solutions.

TECHNICAL SKILLS

Backend & APIs: Java, Spring Boot, REST APIs, microservices, Spring Cloud

Applied AI: Python, local LLMs (Ollama), RAG, semantic search, embeddings, computer vision workflows

Data: SQL, PostgreSQL, pgvector, NumPy, Pandas, relational modeling

Engineering & Delivery: Docker, Git, event-driven communication, Domain-Driven Design (DDD), UML, Scrum

Business & Domain: Requirements gathering, business rules, process and risk analysis, financial operations, Brazil's LGPD data protection law

SELECTED PROJECTS

AI-Academic-Reviewer — Local AI-assisted review pipeline: Designed and implemented a Python/Ollama pipeline that extracts PDF content, analyzes images with a vision model, and generates consolidated Excel reports. Majority voting across runs improves reliability; results were compared with manual reviews.

Legal RAG — Brazilian legislation search and answers: Developed a RAG system for Brazil's Consumer Defense Code, combining legal text structuring, semantic retrieval, and grounded answers from local models with PostgreSQL/pgvector. GitHub: github.com/DiegoOCruz/rag-juridico

RAG from Scratch — Semantic search: Built a pipeline without off-the-shelf RAG frameworks, using NumPy, Ollama, PostgreSQL, and pgvector. Applied it to 41,000 reviews processed with local LLMs, including bottleneck analysis and vector query optimization. GitHub: github.com/DiegoOCruz/estudos-ia

DDD Microservices — Spring Boot and Spring Cloud: Modeled domain logic using DDD, including a ubiquitous language, business rules concentrated in the domain layer, and event-driven communication between services.

Affiliate Deal Automation — n8n: Mapped and automated a manual deal promotion workflow through integrations across platforms. Workflow in production.

PROFESSIONAL EXPERIENCE

Teaching Assistant | Backend Development Course — Instituto Infnet | 2026

- Supported backend development learning activities.
- Developed AI-Academic-Reviewer, a local LLM pipeline that extracts and analyzes PDF and image content and generates consolidated Excel reports.
- Validated the pipeline against manual reviews; in some cases, results were equivalent to or better than manual review.

Founder | MF Colecionáveis | Apr 2025–Present · Part-time

- Created and operate an online collectibles and made-to-order 3D printing store, managing operations from distributor purchasing through sales and after-sales support.
- Generated six-figure revenue in 2026 and completed 500+ sales with no delays or cancellations; earned Mercado Livre's highest seller reputation and an invitation to use Mercado Livre Full.
- Investigated return causes and reduced the rate from 15% to 2% through a packaging and process change at virtually no cost.
- Structured order, inventory, and tax invoicing workflows using Olist.

PROFESSIONAL EXPERIENCE (CONTINUED)

Managing Partner | Zadoc Couros Ltda. | Sep 2009–Present · Penápolis, SP

- Works across several operational areas, currently focused on financial management, at a leather industry company exporting to Nigeria, China, Italy, Vietnam, and other markets.
- Analyzed freight costs and led a change that reduced the rate from BRL 6.50/km to BRL 4.50/km (approximately 31%), while eliminating the company-owned fleet.
- Coordinated end-to-end exports, liaising with international customers, customs brokers, shipping lines, freight forwarders, and banks.
- Defined management reports and metrics, including income statements, cash flow, and forecasts; traceable records enabled a response within 30 minutes to a bank audit of shipments up to two years old.
- Led company restructuring during the pandemic, renegotiating with customers and suppliers and reassessing markets and products.

Attorney | Independent Practice | Jun 2012–Nov 2022

- Practiced corporate, tax, labor, civil, family, and social security law.
- Interviewed clients to identify underlying issues, organize facts, and assess rules and risks before defining a strategy.
- Drafted and reviewed commercial contracts, translating business agreements into clear, verifiable obligations, rules, and conditions.
- Conducted preventive tax planning and analyzed the effects of different corporate structures.

Administrative Assistant | Nova Aliança Couros | Mar 2005–Aug 2009 · Penápolis, SP

- Handled accounts payable and receivable, invoicing, tax controls, and process optimization across departments.

EDUCATION

- B.S. in Software Engineering — Instituto Infnet (2022–2026)
- Bachelor of Laws (LL.B.) — Unitoledo (2011)

COURSES

- English Law — EF International School, Chicago (2013)
- Business English — CCAA (2021)
- Mediation and Arbitration — CAMANP (2011)
- Tax Law — LFG (2011)

LANGUAGES

- Portuguese — Native
- English — Advanced listening and conversation; fluent reading; intermediate writing